



Fair, Accountable, Transparent Machine Learning (FAccT ML)

ZG517



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Session 5
Time: 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Counterfactual Fairness

- Fair Causal Reasoning
- Counterfactual Fairness
 - Formal Methods
 - Law School
 - Crime Rates in NYC
- Equalized Counterfactual Odds
- Multiple Causal Worlds

Readings

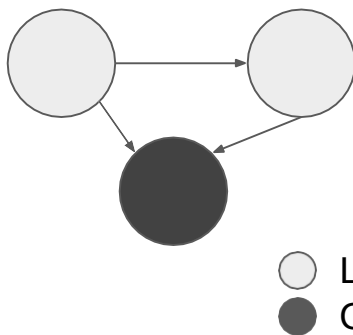


- <https://www.borealisai.com/research-blogs/tutorial1-bias-and-fairness-ai/>
- Chapter 5 of Fairness in Machine Learning Book.
- Matt Kusner et al., Counterfactual Fairness, 2018
- Loftus et al., Causal Reasoning for Algorithmic Fairness, 2018
- Chris Russell et al., When Worlds Collide: Integrating Different Counterfactual Assumptions in Fairness, 2017
- Stephen Pfohl et al., Counterfactual Reasoning for Fair Clinical Risk Prediction, 2019

Fair Causal Reasoning

Causal Graph

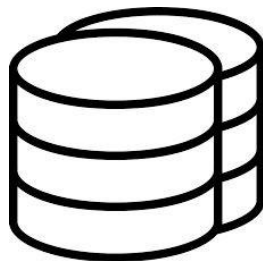
- Observed Data
- Latent Data
- Relations



Causal Fairness Criteria

- Counterfactual Fairness

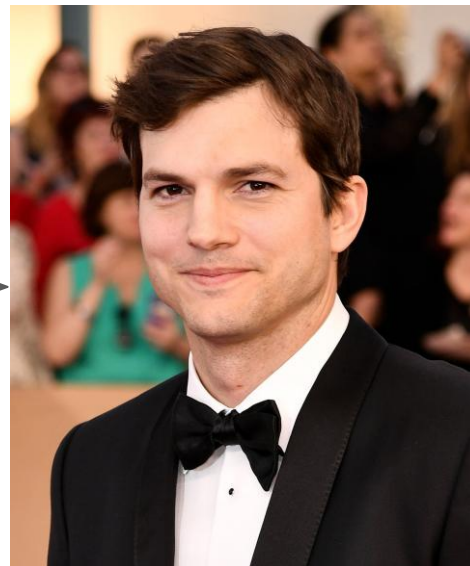
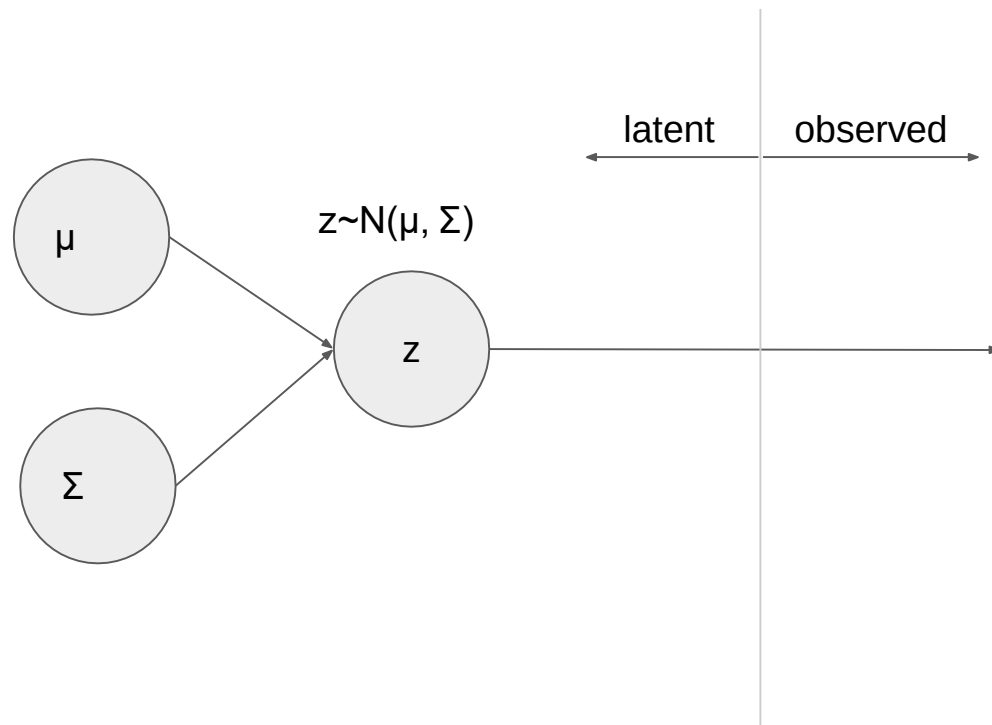
Observed Data



Observational Fairness Criteria

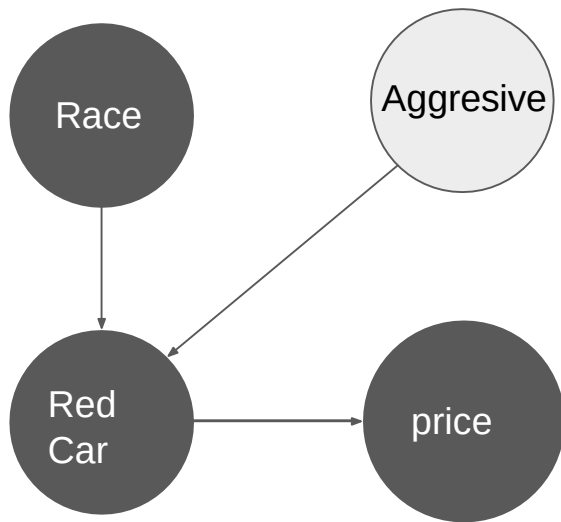
- Fairness Through Unawareness
- Demographic Parity
- Equalized Odds/Opp

Causal Graph



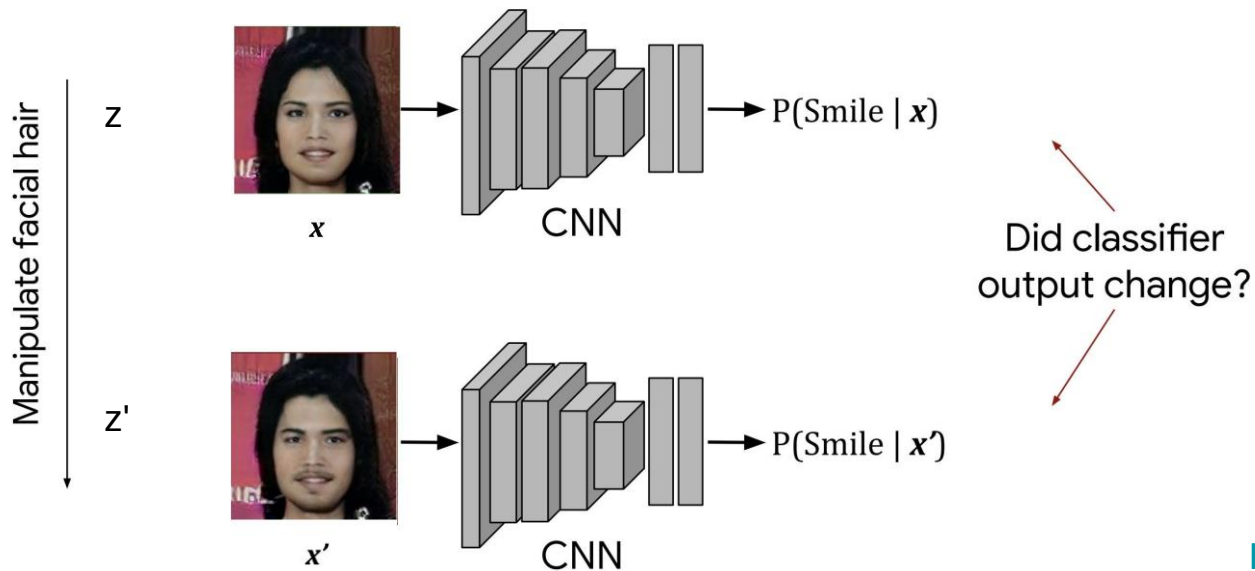
Why Do We Need Causal Fairness?

- Recover Latent Variables



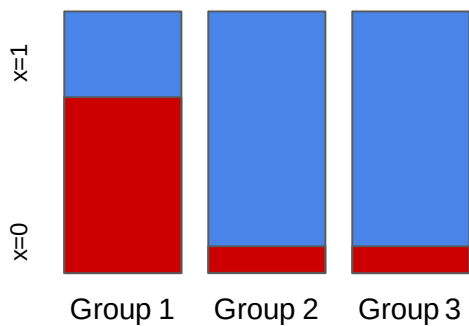
Why Do We Need Causal Fairness?

- Recover Latent Variables

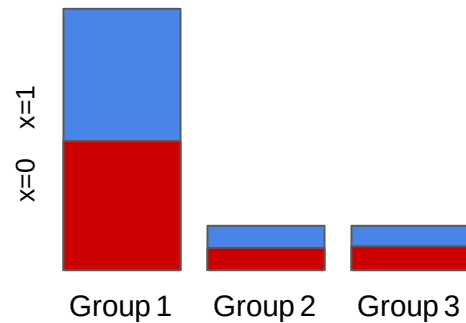


Why Do We Need Causal Fairness?

- Dealing with Inherent bias



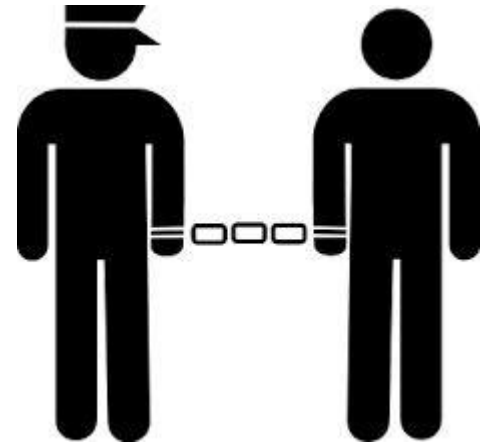
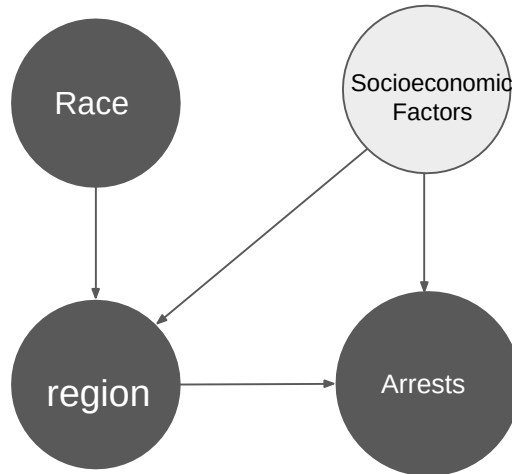
Inherent Biases



Sampling Biases

Inherent bias

- Race groups live in certain regions due to socioeconomic status
- Latent Socioeconomic factors
 - More police resources in regions with low economic status
 - Results in more arrests

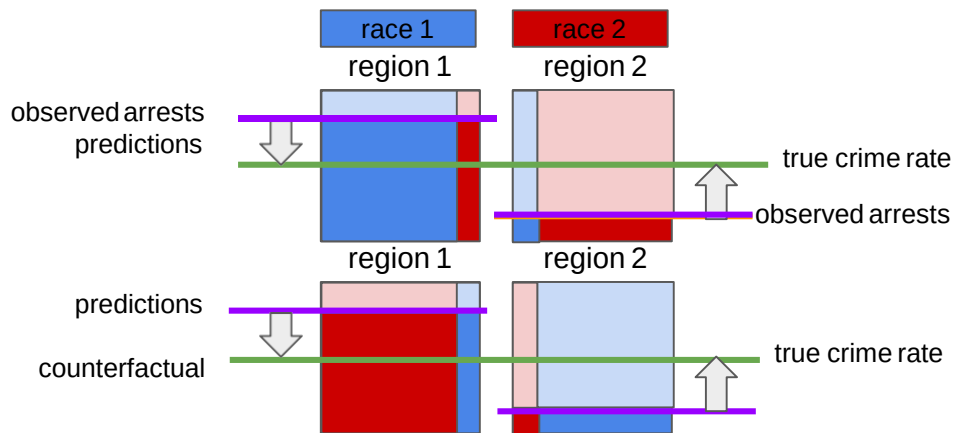


[Kusner et al, 2018](#)

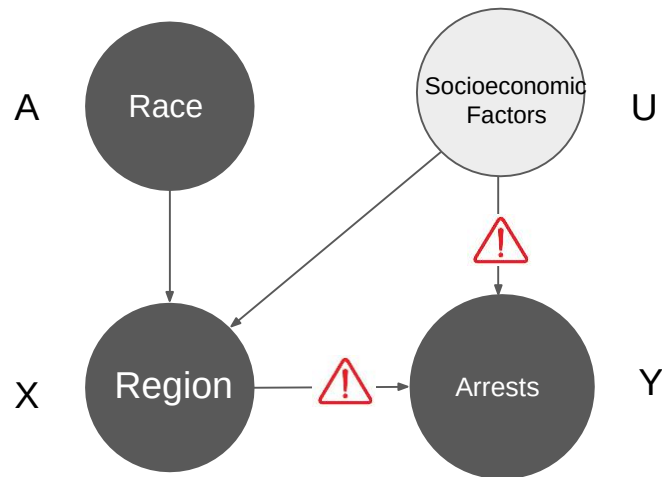
Inherent bias

- Causal Fairness

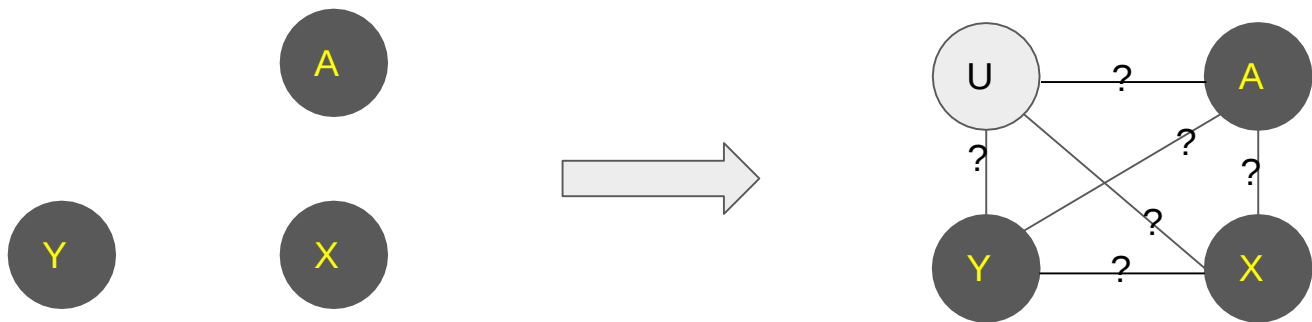
- Intervene variables in a causal graph
- Generating samples with races that live in neighborhood that have high police resources



$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$



Counterfactual Fairness



$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$

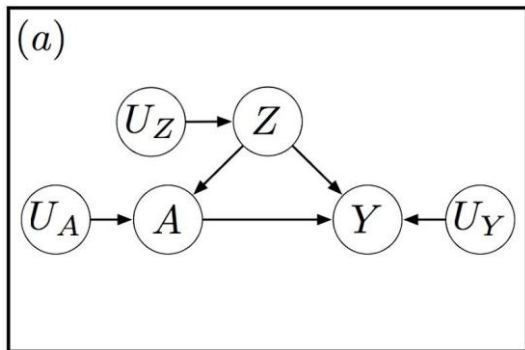
Real Examples

Intervention on $A \leftarrow a$

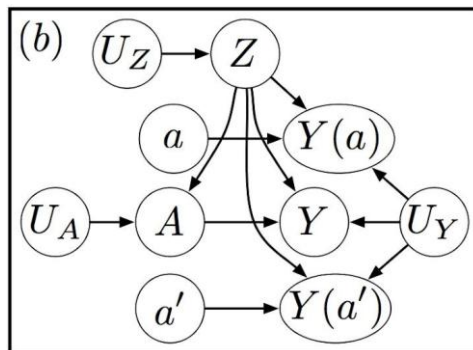
Counterfactual Examples

Intervention on $A \leftarrow a'$

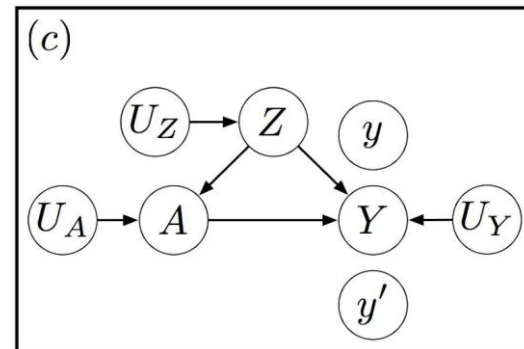
Intervention on Causal Graphs



Causal Graph with A, Z, Y



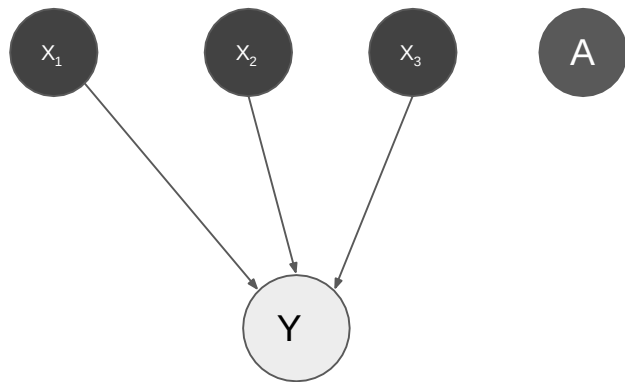
Intervene on A



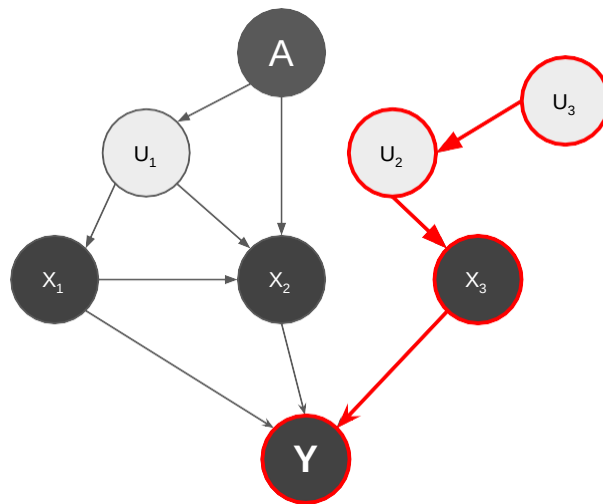
Intervene on Y

Counterfactual Fairness

- Level 1
 - Build predictors using only the observable non-descendants of A

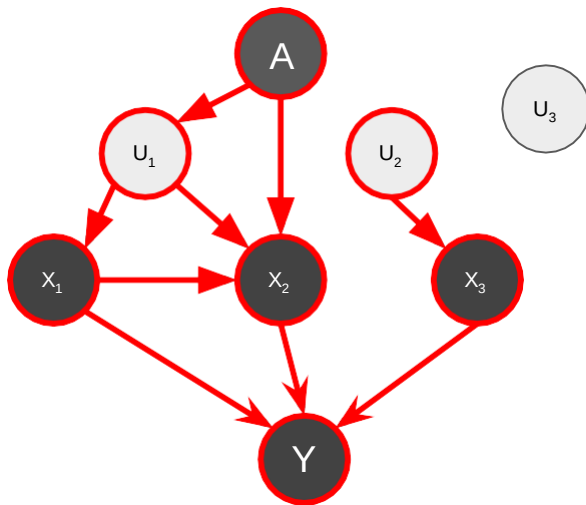


Fairness Through Unawareness



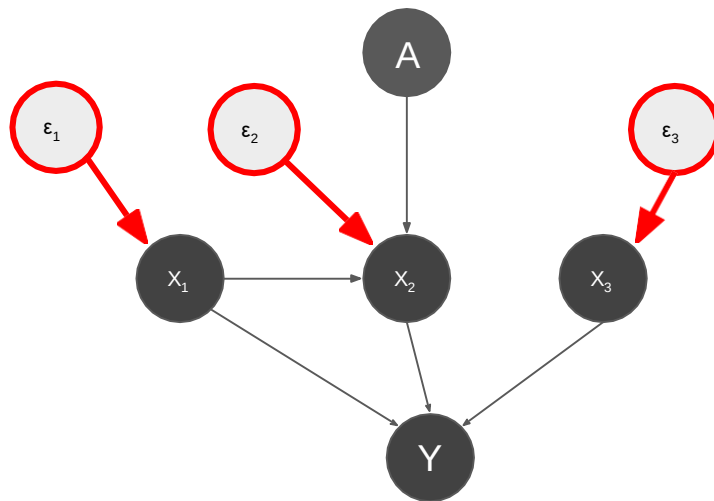
Counterfactual Fairness

- Level 2
 - Build Predictors using the parents of the observable variables



Counterfactual Fairness

- Level 3
 - Build Predictors by adding independent error terms

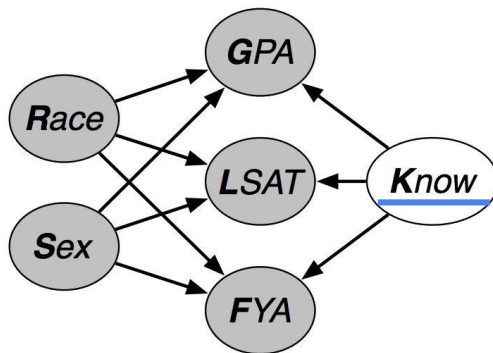


Law School Success Dataset

- Conducted by Law School Admission Council in US
 - 21,790 law students
 - Entrance exam scores (LSAT)
 - Grade-point average (GPA) collected prior to law school
 - Prediction Y = first year average grade (FYA)
 - Protected features = {Gender, Race}

Level 2 Counterfactual Fairness

- Build Predictors using the parents of the observable variables



$$K \sim \mathcal{N}(0, 1)$$

Gaussian Dist. $\swarrow \searrow$

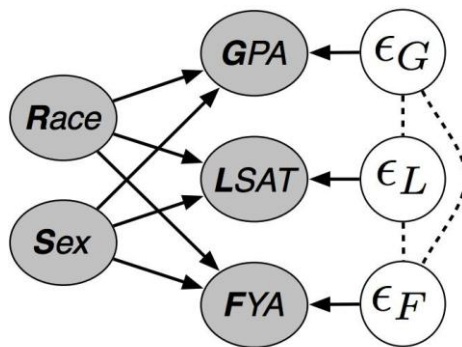
Parameters

$$\text{FYA} \sim \mathcal{N}(\underbrace{w_F^K K}_{\text{blue}} + \underbrace{w_F^R R}_{\text{red}} + \underbrace{w_F^S S}_{\text{red}}, 1) \quad \text{GPA} \sim \mathcal{N}(b_G + w_G^K K + w_G^R R + w_G^S S, \sigma_G)$$
$$\text{LSAT} \sim \text{Poisson}(\exp(b_L + w_L^K K + w_L^R R + w_L^S S))$$

[Kusner et al, 2018](#)

Level 3 Counterfactual Fairness

- Build Predictors by adding independent error terms

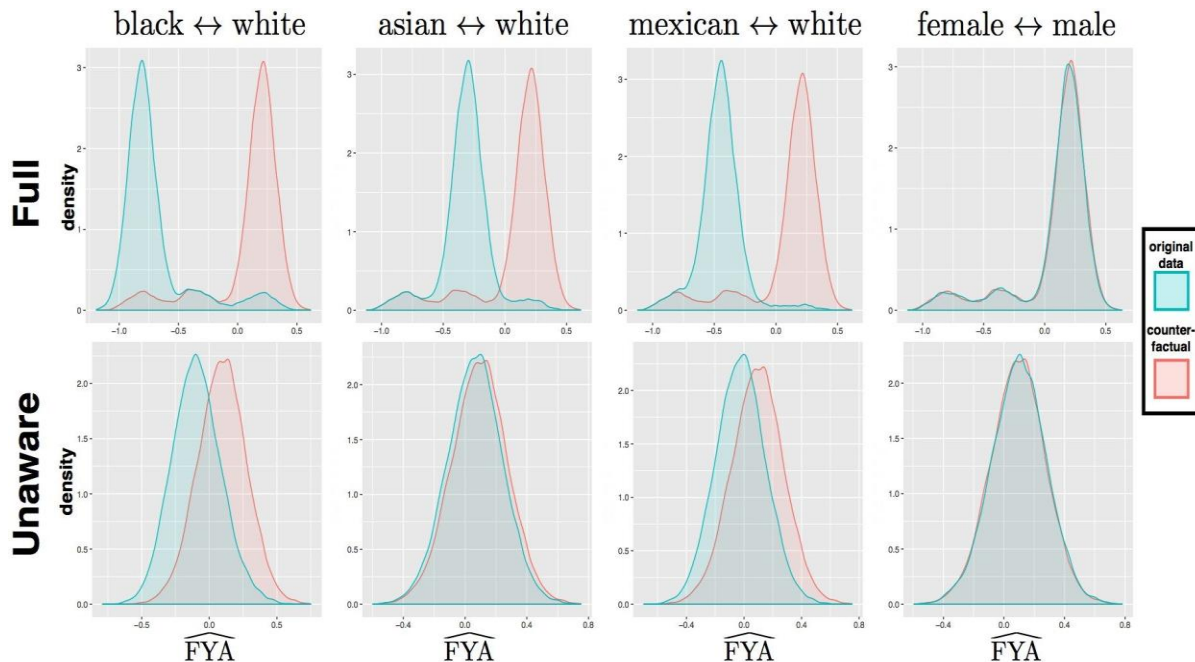


$$\text{GPA} = b_G + w_G^R R + w_G^S S + \epsilon_G, \quad \epsilon_G \sim p(\epsilon_G)$$

$$\text{LSAT} = b_L + w_L^R R + w_L^S S + \epsilon_L, \quad \epsilon_L \sim p(\epsilon_L)$$

$$\text{FYA} = b_F + w_F^R R + w_F^S S + \epsilon_F, \quad \epsilon_F \sim p(\epsilon_F)$$

Baselines



full - using all features

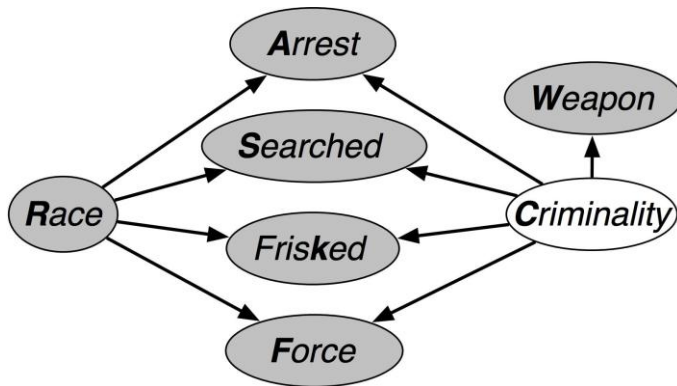
unaware - fairness through unawareness

Results

	Baseline	Baseline	Level 2	Level 3
	Full	Unaware	Fair K	Fair Add
RMSE	0.873	0.894	0.929	0.918

Causal Graph

- Assess the fairness of the NYC arrest dataset
 - 38,609 records
 - White individuals (4492)
 - Black Hispanic individuals (2414)



Assessment Results

White (4492)
Black Hispanic (2414)

White (12.1%)
Black Hispanic (19.8%)

Arrests decreases from
5659 to 3722

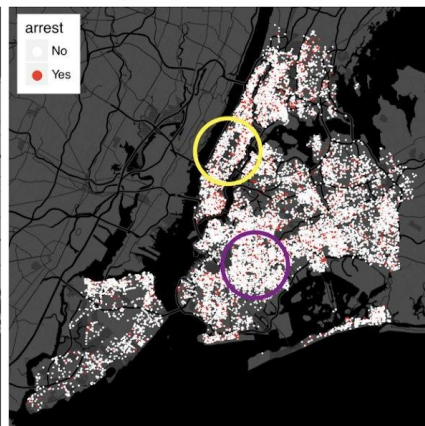
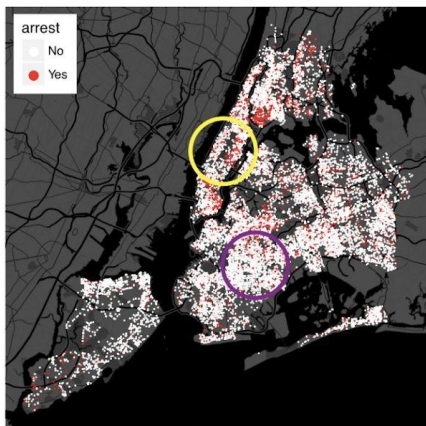
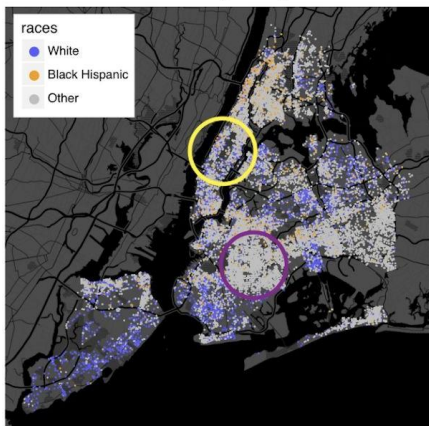
Arrests increases from
5659 to 6439

Race (data)

Arrested (data)

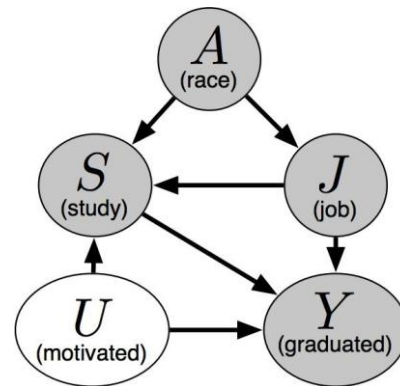
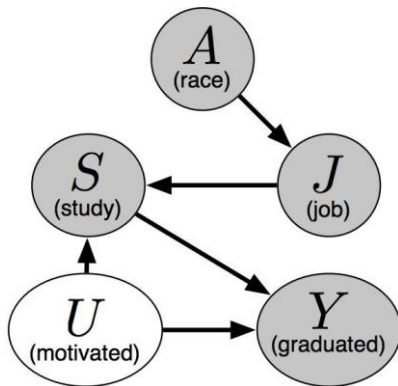
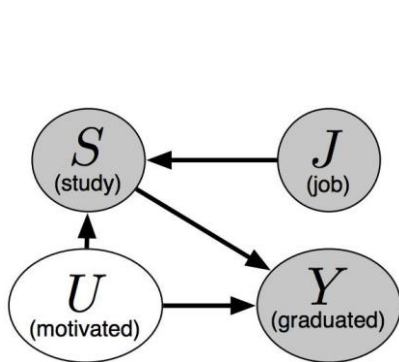
**Arrest if White
(counterfactual)**

**Arrest if Black Hispanic
(counterfactual)**



Multiple Causal Graphs

- Whether a student can graduate on time



Alternative Definitions of Counterfactual Fairness

Exact Formulation

$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$

ϵ - Approximate Formulation

$$\left| f(\mathbf{x}_{A \leftarrow a}, a) - f(\mathbf{x}_{A \leftarrow a'}, a') \right| \leq \epsilon$$

(δ, ϵ) - Approximate Formulation

$$\mathbb{P}(\left| f(\mathcal{X}_{A \leftarrow a}, a) - f(\mathcal{X}_{A \leftarrow a'}, a') \right| \leq \epsilon \mid \mathcal{X} = \mathbf{x}, A = a) \geq 1 - \delta$$

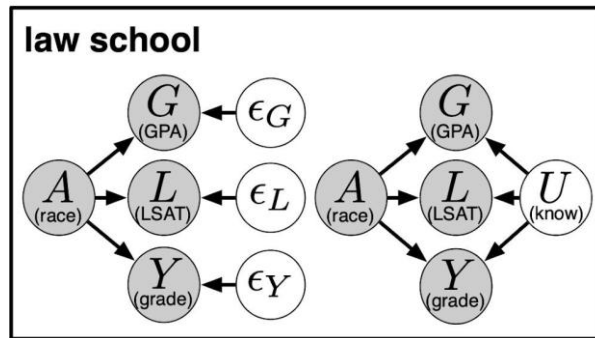
Multi-world Counterfactual Fairness

$$\min_f \frac{1}{n} \sum_{i=1}^n \underbrace{\ell(f(\mathbf{x}_i, a_i), y_i)}_{\text{loss of the data}} + \lambda \sum_{\substack{j=1 \\ \text{world } j}}^m \frac{1}{n} \sum_{i=1}^n \sum_{a' \neq a_i} \underbrace{\mu_j(f, \mathbf{x}_i, a_i, a')}_{\text{counterfactual examples}}$$

$$\mu_j(f, \mathbf{x}_i, a_i, a') := \frac{1}{S} \sum_{s=1}^S \max\{0, \underbrace{|f(\mathbf{x}_{i, A^j \leftarrow a_i}^s, a_i) - f(\mathbf{x}_{i, A^j \leftarrow a'}^s, a')|}_{\epsilon - \text{Approximate Counterfactual Fairness}} - \epsilon\}$$

Monte-carlo Samples

Law Graduate School



$$G = b_G + w_G^A A + \epsilon_G$$

$$L = b_L + w_L^A A + \epsilon_L$$

$$Y = b_Y + w_Y^A A + \epsilon_Y$$

$$\epsilon_G, \epsilon_L, \epsilon_Y \sim \mathcal{N}(0, 1)$$

L3 Method

$$G \sim \mathcal{N}(b_G + w_G^A A + w_G^U U, \sigma_G)$$

$$L \sim \text{Poisson}(\exp(b_L + w_L^A A + w_L^U U))$$

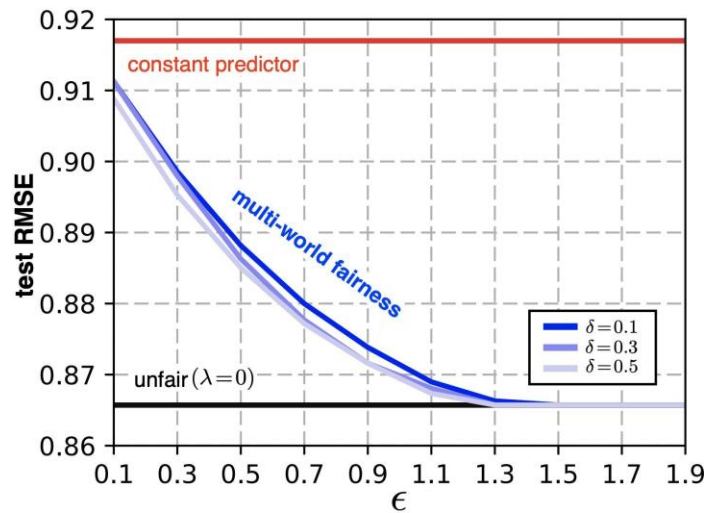
$$Y \sim \mathcal{N}(w_Y^A A + w_Y^U U, 1)$$

$$U \sim \mathcal{N}(0, 1)$$

L2 Method

[Russell et al, 2017](#)

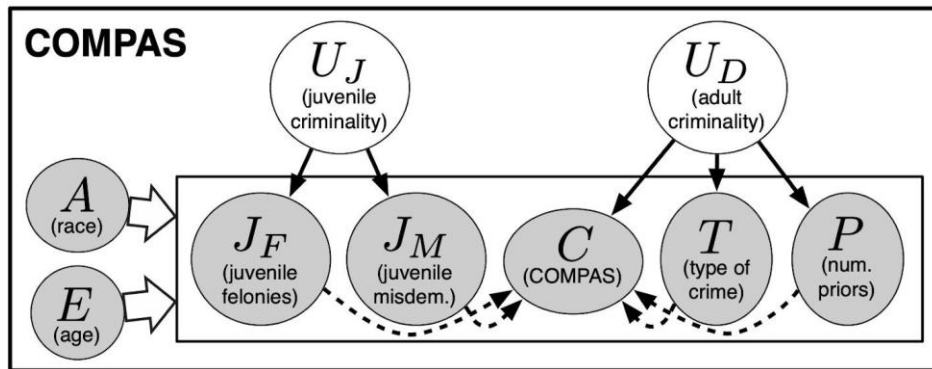
Results



$$|f(\mathbf{x}_{A \leftarrow a}, a) - f(\mathbf{x}_{A \leftarrow a'}, a')| \leq \epsilon$$

[Russell et al, 2017](#)

COMPAS



$$T \sim \text{Bernoulli}(\phi(b_T + w_C^{U_D} U_D + w_C^E E + w_C^A A)) \quad ($$

$$C \sim \mathcal{N}(b_C + w_C^{U_D} U_D + w_C^E E + w_C^A A + w_C^T T + w_C^P P + w_C^{J_F} J_F + w_C^{J_M} J_M, \sigma_C)$$

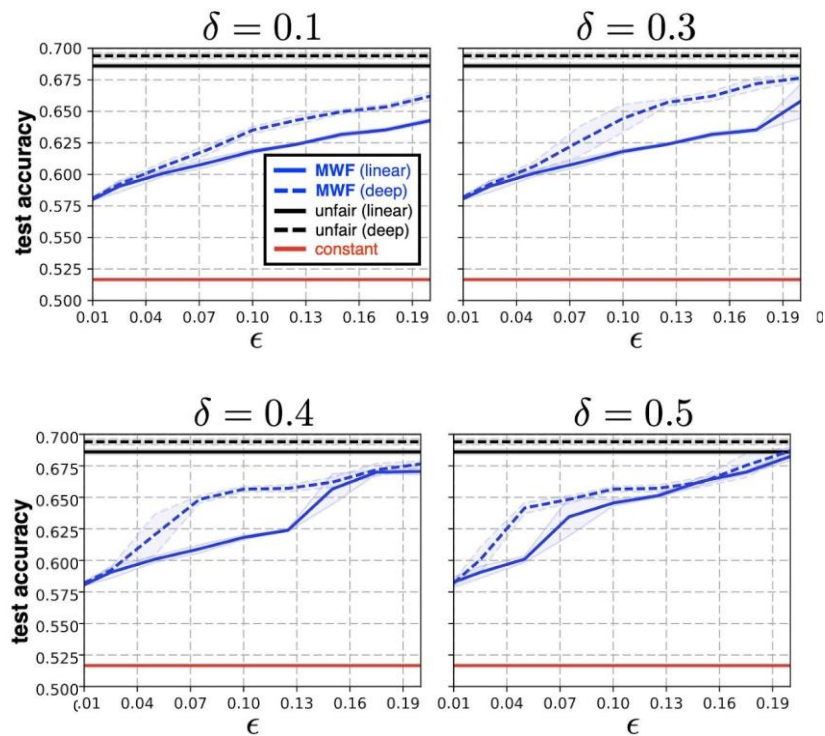
$$P \sim \text{Poisson}(\exp(b_P + w_P^{U_D} U_D + w_P^E E + w_P^A A))$$

$$J_F \sim \text{Poisson}(\exp(b_{J_F} + w_{J_F}^{U_J} U_J + w_{J_F}^E E + w_{J_F}^A A))$$

$$J_M \sim \text{Poisson}(\exp(b_{J_M} + w_{J_M}^{U_J} U_J + w_{J_M}^E E + w_{J_M}^A A))$$

$$[U_J, U_D] \sim \mathcal{N}(0, \Sigma)$$

Results



[Russell et al, 2017](#)

Equalized Counterfactual Odds

Equality of Odds

$$P(\hat{Y} = 1 | A = 0, Y) = P(\hat{Y} = 1 | A = 1, Y)$$

Counterfactual Fairness

$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$

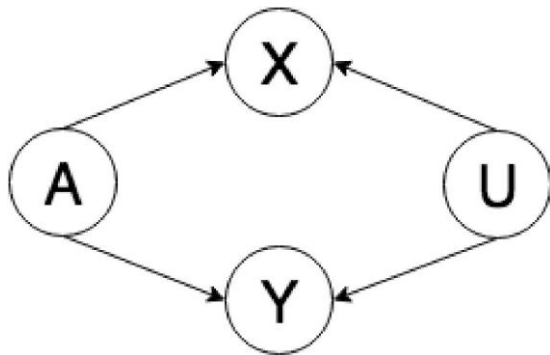
Equalized Counterfactual Odds

$$p(\hat{Y}_{A \leftarrow a}(U) \mid X = x, Y_{A \leftarrow a} = y, A = a) = p(\hat{Y}_{A \leftarrow a'}(U) \mid X = x, Y_{A \leftarrow a'} = y, A = a)$$

- “the predictor is counterfactually fair, conditioned on the factual outcome Y matching the counterfactual outcome $Y_{A \leftarrow a}$.”
- In contrast, the original counterfactual fairness formulation can be interpreted as requiring predictions to be the same across factual counterfactual pairs, regardless of whether those pairs share the same value of the outcome.”

Healthcare Equality

- Protected Features $A = \{\text{Gender}\}$
- Features X , vector representation of coded diagnoses, procedures, medication orders, lab results, and clinical notes
- Prediction Y , a binary indicator of the occurrence of a clinically relevant outcome



$$u \sim p(U) = \text{Normal}(0, I)$$

$$a \sim p(A) = \text{Categorical}(A \mid \pi)$$

$$x, y \sim p(X, Y \mid U, A) = p(X \mid U, A)p(Y \mid U, A)$$

<https://arxiv.org/pdf/1907.06260>

Training Objective

- σ - sigmoid function
- h - predictor
- J - cross entropy loss

$$\mathcal{L} = J(h_{\theta}(x, a), y) + \lambda_{\text{CF}} \sum_{a_k \in \mathcal{A}} \mathbb{1}[a \neq a_k] J(h_{\theta}(\underbrace{x_{A \leftarrow a_k}}_{\text{logits}}, \underbrace{a_k}_{\text{logits}}), \underbrace{y_{A \leftarrow a_k}}_{\text{logits}}) +$$
$$\lambda_{\text{CLP}} \sum_{a_k \in \mathcal{A}} \mathbb{1}[a \neq a_k] \mathbb{1}[y = \underbrace{y_{A \leftarrow a_k}}_{\text{logits}}] \left(\underbrace{\sigma^{-1}(h_{\theta}(\underbrace{x_{A \leftarrow a_k}}_{\text{logits}}, \underbrace{a_k}_{\text{logits}}))}_{\text{logits}} - \underbrace{\sigma^{-1}(h_{\theta}(x, a))}_{\text{logits}} \right)^2$$

- Counterfactual Logit Pairing (CLP) is a technique that addresses such issues to ensure that a model's prediction doesn't change when a sensitive attribute referenced in an example is either removed or replaced. It improves a model's robustness to such perturbations, and can positively influence a model's stability, fairness, and safety.

Dataset Overview

Group	Count	Length of Stay ≥ 7 Days	Inpatient Mortality
Asian	17,465	0.187	0.025
Black	5,202	0.239	0.020
Hispanic	21,978	0.196	0.019
Other	11,004	0.200	0.022
Unknown	3,593	0.201	0.072
White	70,391	0.204	0.021
Female	72,556	0.167	0.018
Male	57,076	0.245	0.029
[18, 30)	15,291	0.180	0.007
[30, 45)	27,155	0.140	0.007
[45, 65)	43,529	0.222	0.025
[65, 89)	43,658	0.226	0.036
All	129,633	0.201	0.023

Results

Group	Metric	λ_{CLP}					
		N/A	0.0	0.01	0.1	1.0	10.0
Asian	AUC-PRC	0.605	0.563	0.555	0.561	0.56	0.562
	AUC-ROC	0.86	0.853	0.853	0.854	0.849	0.851
	Brier	0.106	0.11	0.109	0.109	0.11	0.112
Black	AUC-PRC	0.579	0.548	0.55	0.545	0.563	0.573
	AUC-ROC	0.838	0.825	0.82	0.825	0.823	0.823
	Brier	0.124	0.135	0.129	0.128	0.127	0.129
Hispanic	AUC-PRC	0.592	0.558	0.565	0.57	0.564	0.56
	AUC-ROC	0.862	0.855	0.856	0.861	0.853	0.854
	Brier	0.113	0.117	0.115	0.114	0.117	0.118
Other	AUC-PRC	0.549	0.557	0.557	0.563	0.553	0.561
	AUC-ROC	0.824	0.827	0.819	0.824	0.819	0.827
	Brier	0.122	0.124	0.121	0.121	0.122	0.124
Unknown	AUC-PRC	0.675	0.616	0.616	0.606	0.614	0.633
	AUC-ROC	0.9	0.891	0.888	0.893	0.891	0.887
	Brier	0.104	0.106	0.103	0.103	0.105	0.111
White	AUC-PRC	0.575	0.568	0.564	0.559	0.562	0.563
	AUC-ROC	0.847	0.84	0.839	0.838	0.838	0.837
	Brier	0.118	0.12	0.118	0.12	0.12	0.121

Results

- Difference in the counterfactual versus factual predicted probability

